SIGNAL & DATA ANALYSIS IN NEUROSCIENCE 2020
SPIKE TRAIN ANALYSIS

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Outline

- Analysis of neuronal spike train
  - Hazard function
  - Auto correlation
  - Cross correlation

Single spike train analysis

Term	Definition
ISI	Time difference between adjacent spikes
TIH	Histogram of the ISI
Survivor function	The probability of surviving at least to time t. $Survivor(t) = 1 - CDF\big(ISI(t)\big) = 1 - \sum_{i=1}^{t} ISI(i)$
Hazard function	The probability to fire a spike after we have not fire till now from the previous spike $Hazard(t) = \frac{ISI(t)}{Survivor(t)}$

Properties of the Hazard function

- $h(t) \geq 0$
- $h(t)$ is not a probability, so $\int h(t) \to \infty$
- $h(t)$ for a Poisson distribution =

  probability for (1st) spike at any given time =

  the Poisson spike rate, *r*.
- Noisy for long ISIs.
- For discrete distributions we define the hazard function as:

$$H(t) = \frac{ISI(t)}{1 - CDF(t - \epsilon)}$$

Example: discrete Poisson distribution

- A neuron fires with probability P(t) given by

$$P(t) = \begin{cases} p & t = n \cdot T \\ 0 & \text{otherwise} \end{cases}$$

Draw the survivor and hazard functions

Solution

- $P(ISI = nT) = (1-p)^{n-1} \cdot p$

- $CDF(nT) = \sum_{i=1}^{n} p(ISI = iT)$

$$= p \cdot (1-p)^{1-1} + p \cdot (1-p)^{2-1} + p \cdot (1-p)^{3-1} + \ldots$$

$$= p \cdot 1 \qquad + p \cdot (1-p)^{1} + p \cdot (1-p)^{2} + \ldots$$

$$a + ar + ar^2 + ar^3 + \cdots + ar^{n-1} = \sum_{k=0}^{n-1} ar^k = a\frac{1-r^n}{1-r},$$

$$p\frac{1-(1-p)^n}{1-(1-p)} = 1 - (1-p)^n$$

$CDF(nT) = 1 - P(\text{no spike in first n slots}) = 1 - (1-p)^n$

- $S(nT) = 1 - CDF(nT) = (1-p)^n$

- $H(nT) = P(ISI = nT) / S(nT - \varepsilon) = (1-p)^{n-1} \cdot p / (1-p)^{n-1} = p$

Simulation (p=0.3)

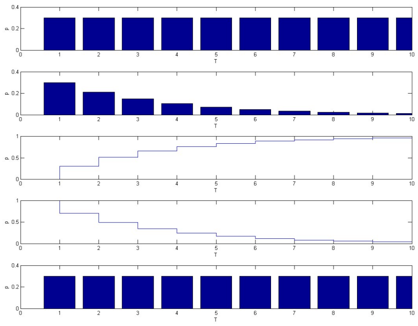
Spike probabilities

ISI pdf(τ)

CDF(τ)

Survivor(τ)
(1-CDF)

Hazard(τ)



Auto (and cross-) correlation definition

Autocorrelation Definition (notation):	$Q(x,y)[n] = \sum_{i=-\infty}^{\infty} x[i] \cdot y[i+n]$

- We denote Q(x,y) as Q(const reference, moving target).
  *Matlab's notation is reversed

Normalization	Definition	Remarks
Count	$Q_C(x,y)[n] = Q(x,y)[n] = \sum_{i=-\infty}^{\infty} x[i] \cdot y[i+n]$	Counts the # of spikes that follow a spike by n slots(Δt)
Probability	$Q_{prob}(x,y)[n] = \frac{1}{\#\text{ of spikes in } x} Q_C(x,y)[n]$	Probability of finding a spike, that follows a spike by n slots (Δt)
Rate	$Q_{rate}(x,y)[n] = \frac{1}{\Delta t} Q_{prob}(x,y)[n]$	Average rate of spikes in a bin, that follows a spike by n Δt.

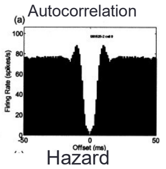
Matlab usage and basic properties

- Matlab usage:
  - [vec] = xcorr(x,x, maxlag)
  - It is partical to zero the middle bin: vec(maxlag +1) =0;
- Properties:
  - xcorr(X,X) is symmetric.

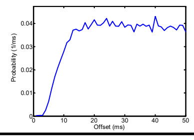
Autocorrelation examples

Poisson with refractory period

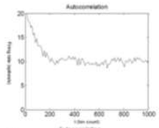
Autocorrelation



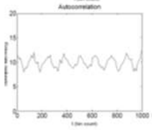
Hazard



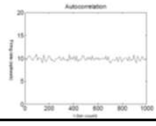
Burster



Oscillator

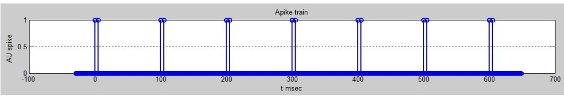


Poisson



Example: Autocorrelation

- Neurons in the Doubelitis Exactis nucleus of the Mountain Troll fire pairs of spikes (5 ms apart) every 100msecs.
  - Calculate the Fano factor and Coefficient of variation of the neurons. Explain the results and their relation to a Poissonian neuron firing at the same rate.
  - Draw the autocorrelation and hazard functions of the neurons (the functions should be drawn in the range 0-250msec).



Solution

- FF: # of spikes over a section is almost constant (window size…) => FF $\rightarrow 0$

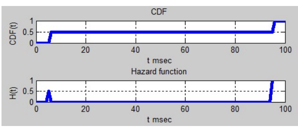
$$\mathrm{FF} = \frac{var[spikeoverT]}{E[spikeoverT]}$$

- Cv: $\mathrm{P(ISI = 5) = P(ISI = 95) = 0.5}$

$$C_v = \frac{\mathrm{std\ (ISI)}}{\mathrm{E(ISI)}}$$

- Hazard(t):

$$H(t) = \frac{ISI(t)}{1-CDF(t)}$$



Cross-correlation: definition

Definition	$Q(x,y)[n] = \sum_{i=-\infty}^{\infty} x[i] \cdot y[i+n]$	$Q(x,y)[n] = \sum_{i=-\infty}^{\infty} x[i+n] \cdot y[i]$
	Peter Dayan	Matlab

- We denote Q(x,y) as Q(const reference, moving target), as per Peter Dayan.

Normalization	Definition	Remarks
Count	$Q_C(x,y)[n] = Q(x,y)[n] = \sum_{i=-\infty}^{\infty} x[i] \cdot y[i+n]$	Counts the # of y spikes that follow a x spike by n Δt.
Probability	$Q_{prob}(x,y)[n] = \frac{1}{\text{\# of spikes in } x} Q_C(x,y)[n]$	Probability of finding a y spike, that follows x spike by n Δt.
Rate	$Q_{rate}(x,y)[n] = \frac{1}{\Delta t} Q_{prob}(x,y)[n]$	Average rate of y in a bin, that follows x spike by n Δt.

Practicalities: Auto- vs. Cross- Correlation

Property	**Auto-correlation**	**Cross-correlation**
Definition	$Q(x,x)[n] = \sum_{i=-\infty}^{\infty} x[i] \cdot x[i+n]$	$Q(x,y)[n] = \sum_{i=-\infty}^{\infty} x[i] \cdot y[i+n]$ Q(ref, target)
Symmetry	Symmetric	Not Symmetric
Matlab usage	xcorr(x,x, maxlag); vec(maxlag +1) = 0	**fliplr**(xcorr(x,y,maxlag));

Matlab: "Step" forward on X (reference) vector
Dayan (and us): "Step" forward on Y (target) vector
Result: Mirror image.
Therefore:

1. Use matlab xcorr.
2. "reverse" (e.g fliplr([1 2 3 4 5]) = [5 4 3 2 1];)

Example: regular neurons

- Neuron A fires regularly at 10spike/sec,
  t = 0, 100, 200, 300, 400, … msec.
  Following each spike of A: neuron B fires 10 spikes at 1msec intervals with initial delay of 20ms.
  e.g: 20, 21, …29, 120, 121, msec etc.
  - Draw example spike-trains of A and B
  - Draw the cross correlation of A->B normalized to probability.
  - Draw the cross correlation of B->A normalized to probability.

Example: regular n. activates Poisson n.

- Neuron A sends excitatory inputs to neurons B and C (B and C are not directly connected).
  - Neuron A fires regularly (a perfect pacemaker) at 5spikes/s
  - Neuron B is a Poisson process with a rate of 60 spikes/s increasing to 80 spikes/s for a period of 5ms following an incoming spike from neuron A.
  - Neuron C is a Poisson process with a rate of 0 spikes/s increasing to 20 spikes/s for a period of 10ms following an incoming spike from neuron A.
  - Draw the cross-correlation of A-B, A-C, B-C and C-B (in the range ±500ms).